

Introduction to Probabilistic Machine Learning

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Graphical Models: Independence

Overview

1. Graphical Models
2. Bayesian Networks
3. Conditional Independence

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*Unit 3 – Graphical Models:
Independence*

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Graphical Models

- **Challenge:** How to formulate complex likelihoods/data models & priors for *actual* data?

- **Example 1:** Match outcomes $y \in \{-1, 1\}$ (data) for a head-to-head match between two players

- **Prior:** $p(\mathbf{s}) = \mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2)$ ← skill belief

- **Likelihood:** $p(y|\mathbf{s}) = \int \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0) dp_1 dp_2$ ← marginalization

Match outcome

Player performance

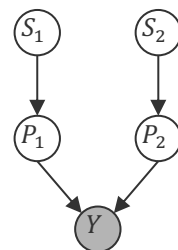
- **Observation:** The product structure of the probabilities seems crucial

- **Idea:** Define a graph where each of the variables are nodes and edges indicate factor relationships between variables

- **Advantages:** Simple way to visualize factor structure of the joint probability

- **Graphical Models:** Conditional independence based on graph properties

- **Factor Graphs:** Efficient inference and approximation algorithms



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Bayesian Networks

- **Observation.** Any joint distribution $p(x_1, \dots, x_n)$ can be written as

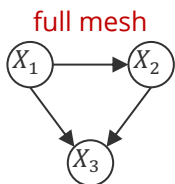
$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | x_1, \dots, x_{i-1})$$

- **Bayesian Network.** Given a joint distribution

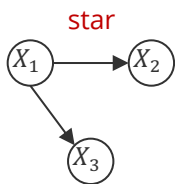
$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \mathbf{x}_{\text{parents}_i})$$

a Bayesian network is a graph with a node for every variable X_i , and a directed edge from every variable $X_j, j \in \text{parent}_i$ to X_i .

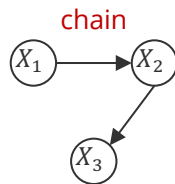
- A Bayesian network must always be a **directed acyclic graph** because only those have a topological order corresponding to a variable order.



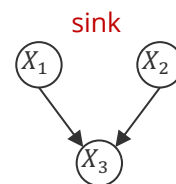
$$p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2 | x_1) \cdot p(x_3 | x_1, x_2)$$



$$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^3 p(x_i | x_1)$$



$$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^3 p(x_i | x_{i-1})$$



$$p(x_1, x_2, x_3) = p(x_3 | x_1, x_2) \cdot \prod_{i=1}^2 p(x_i)$$

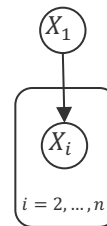
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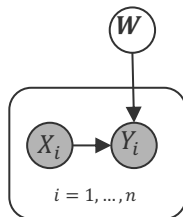
Bayesian Network (Notation)

- **Plate.** If a subset of variables has the same relation only differing in their index, we use a "plate" to collapse them into a single graphical element.
 - Increase readability of models for large amounts of parameters and data
- **Observed Variables.** If a subset of variables has been observed ("data"), the variable nodes are usually shaded ("clamped").
 - **Example:** Discriminatory Models

$$p(x_1, x_2, \dots, x_n) = p(x_1) \cdot \prod_{i=2}^n p(x_i | x_1)$$



$$p(\mathbf{w}, y_1, \dots, y_n | x_1, \dots, x_n) = \prod_{i=1}^n p(y_i | x_i, \mathbf{w}) \cdot p(\mathbf{w})$$



$$p(\mathbf{w} | (x_1, y_1), \dots, (x_n, y_n)) \propto \prod_{i=1}^n p(y_i | x_i, \mathbf{w}) \cdot p(\mathbf{w})$$

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Representation Complexity

- For simplicity, let us assume that $x_i \in \{1, \dots, K\}$

Naive

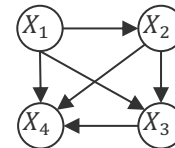
$$p(x_1, \dots, x_n)$$

x_1	x_2	x_3	x_4	$p(x_1, x_2, x_3, x_4)$
1	1	1	1	p_{1111}
1	1	1	2	p_{1112}
⋮				
1	1	1	K	p_{111K}
1	1	2	1	p_{1121}
⋮				
K	K	K	K	$1 - \Sigma$

$$K^4 - 1$$

Bayesian Network

$$p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2) \cdot p(x_4|x_1, x_2, x_3)$$



x_1	$p(x_1)$
1	p_1
2	p_2
⋮	
K	$1 - \Sigma$

$$K - 1$$

x_2	x_1	$p(x_2 x_1)$
1	1	p_{11}
2	1	p_{21}
⋮		
K	1	$1 - \Sigma$
1	2	p_{12}
⋮		
K	K	$1 - \Sigma$

$$(K - 1) \cdot K$$

x_3	x_1	x_2	$p(x_3 x_1, x_2)$
1	1	1	p_{111}
2	1	1	p_{211}
⋮			
K	1	1	$1 - \Sigma$
1	1	2	p_{112}
⋮			
K	K	K	$1 - \Sigma$

$$(K - 1) \cdot K^2$$

x_4	x_1	x_2	x_3	$p(x_4 x_1, x_2, x_3)$
1	1	1	1	p_{1111}
2	1	1	1	p_{2111}
⋮				
K	1	1	1	$1 - \Sigma$
1	1	1	2	p_{1112}
⋮				
K	K	K	K	$1 - \Sigma$

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$$(K - 1) \cdot K^3$$

$$\begin{aligned}
 (K - 1) \cdot (1 + K + K^2 + K^3) &= (K + K^2 + K^3 + K^4) - (1 + K + K^2 + K^3) \\
 &= K^4 - 1
 \end{aligned}$$

Sampling a Bayesian Network

- One advantage of a Bayesian network is the ability to *sample* $p(x_1, \dots, x_n)$

Ancestral Sampling

1. Topologically sort all variables $X_1, \dots, X_n$ into $X_{(1)}, \dots, X_{(n)}$
2. Sample each variable $X_{(i)}$ using distribution $p(X_{(i)} | X_{(1)}, \dots, X_{(i-1)})$

- **Assumption**

1. Sampling from the conditional distributions is simpler than from the joint distribution
2. There are no clamped nodes, that is, we do not condition on any variable

- **Problems**

1. Sampling is *sequential* one variable at the time
2. Conditioning happens only on frequent events because for samples $x_{j,1}, \dots, x_{j,n}$

$$\frac{|\{(x_{j,1}, \dots, x_{j,n}) \mid x_{j,1} = x_1 \wedge \dots \wedge x_{j,n} = x_n\}|}{|\{x_{j,n} = x_n\}|} \approx \frac{p(x_1, \dots, x_n)}{p(x_n)} = p(x_1, \dots, x_{n-1} | x_n)$$

≈ 0 for rare events

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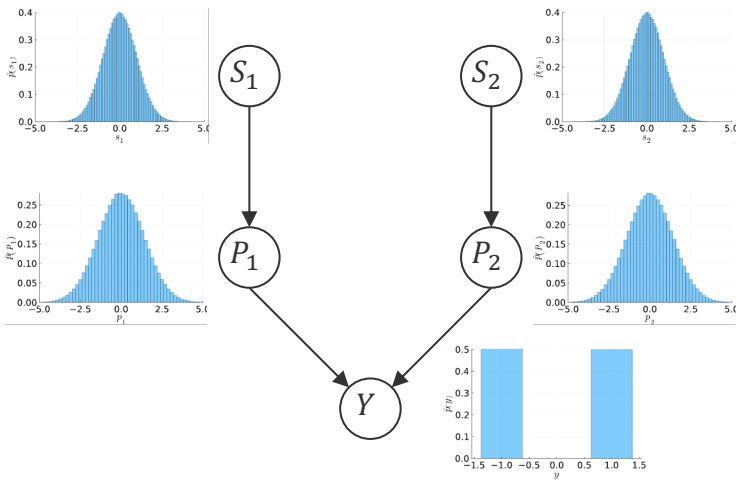
Sampling a Bayesian Network: Example

$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$

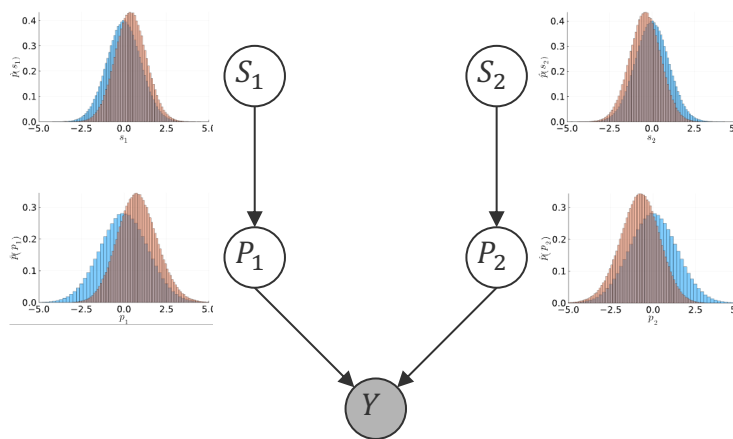
$$\mu_1 = \mu_2$$

```
# samples from the TrueSkill graphical model
function sample(; n = 100000, μ1=0.0, σ1=1.0, μ2=0.0, σ2=1.0, β=1.0)
    samples = Vector{Float64}{}(undef, n)
    for i in 1:n
        s1 = rand(Normal(μ1, σ1))
        s2 = rand(Normal(μ2, σ2))
        p1 = rand(Normal(s1, β))
        p2 = rand(Normal(s2, β))
        y = p1 > p2 ? 1.0 : -1.0
        samples[i] = [s1, s2, p1, p2, y]
    end
    return samples
end
```

Without match outcome



With match outcome ($y = 1$)



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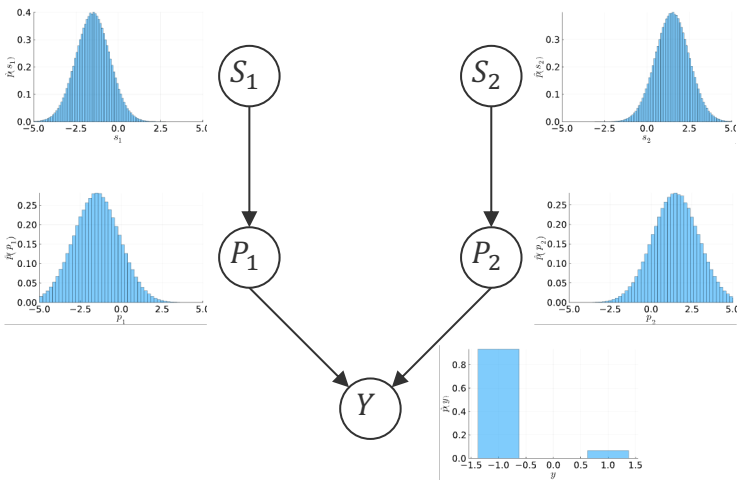
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Sampling a Bayesian Network: Example (ctd)

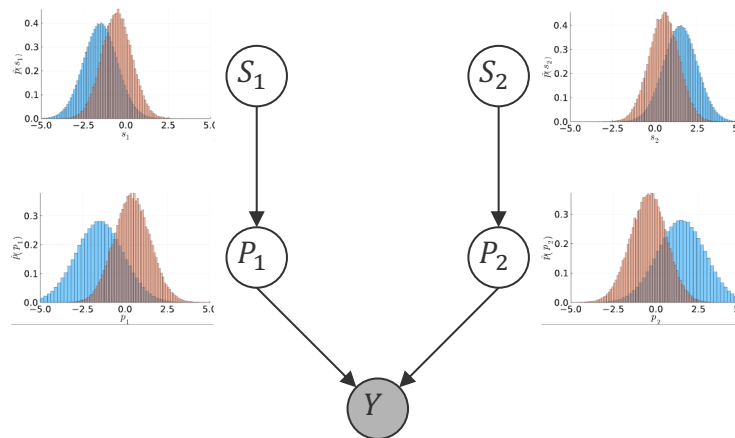
$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$

$$\mu_1 \ll \mu_2$$

Without match outcome



With match outcome ($y = 1$)



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Conditional Independence

- In modelling specific data, domain experts often know whether two (latent) measurements can affect each other or not (i.e., are independent)
 - **Examples:**
 - Skills of two players in a video game are not dependent if they never played before
 - Skills of two players in a video game *are* dependent if they have played many times!



Philip Dawid
(1946–)

- Bayesian networks are useful to determine conditional independence.
- **Conditional Independence.** A random variable X_i is conditionally independent of a random variable X_j given the variable X_k if for all values X_k

$$p(X_i = x_i | X_j = x_j, X_k = x_k) = p(X_i = x_i | X_k = x_k)$$

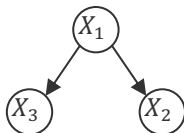
- Equivalent definition: $p(X_i = x_i, X_j = x_j | X_k = x_k) = p(X_i = x_i | X_k = x_k) \cdot p(X_j = x_j | X_k = x_k)$
- Shorthand notation (Dawid, 1979): $X_i \perp X_j | X_k$

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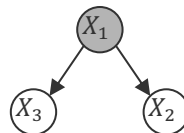
Conditional Independence: Warm-Up I

- **Tail-to-Tail Node (x_1):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)$



$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1) \neq p(x_2) \cdot p(x_3)$$

not independent



$$p(x_2, x_3|x_1) = \frac{p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)}{p(x_1)} = p(x_2|x_1) \cdot p(x_3|x_1)$$

conditionally independent

- **Head-to-Tail Node (x_2):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_2)$



$$p(x_1, x_3) = p(x_1) \cdot \sum_{\{x_2\}} p(x_2|x_1) \cdot p(x_3|x_2) = p(x_1) \cdot p(x_3|x_1) \neq p(x_1) \cdot p(x_3)$$

not independent



$$p(x_1, x_3|x_2) = \frac{p(x_2|x_1) \cdot p(x_1)}{p(x_2)} \cdot p(x_3|x_2) = p(x_1|x_2) \cdot p(x_3|x_2)$$

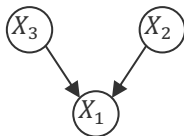
conditionally independent

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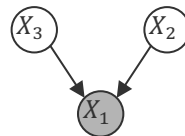
Conditional Independence: Warm-Up II

- **Head-to-Head Node (x_1):** $p(x_1, x_2, x_3) = p(x_2) \cdot p(x_3) \cdot p(x_1|x_2, x_3)$



$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3) = p(x_2) \cdot p(x_3)$$

independent

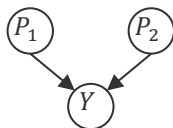


$$p(x_2, x_3|x_1) = \frac{p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3)}{p(x_1)} \neq p(x_2|x_1) \cdot p(x_3|x_1)$$

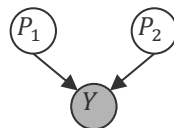
not (always) conditionally independent

- It can be shown that the path between X_2 and X_3 are only independent if *none* of the *descendant* node from X_1 (that can be reached in the directed graph) is observed!

- **Skill Example (ctd):** Consider the performance of two players



Before match: P_1 and P_2 are independent



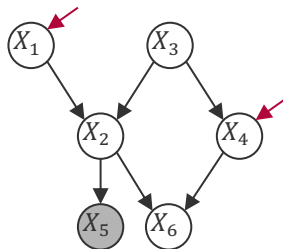
After match: P_1 and P_2 are **not** independent

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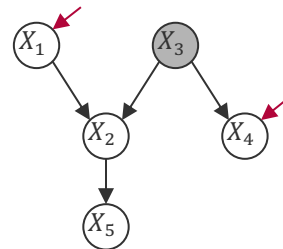
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Conditional Independence: d-separation

- **Blocked Node.** A node in a Bayesian network is said to be blocked if
 1. It's a head-to-tail or tail-to-tail node and the node is observed.
 2. It's a head-to-head node and neither the node nor any of its descendants are observed.
- **d-separation.** Given a Bayesian network and a subset of observed variables, two non-observed variables X_i and X_j are conditionally independent (that is, d-separated) if every undirected path between X_i and X_j contains at least one blocked node.
- **Examples.**



X_1 and X_4 are not independent



X_1 and X_4 are independent



Judea Pearl
(1936–)

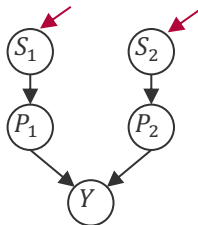
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Conditional Independence: Skill Example

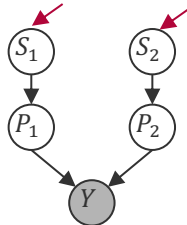
- **Skill Example (ctd):** Consider the skills of two players

Before match



S_1 and S_2 are independent

After match



S_1 and S_2 are **not** independent

- Intuitive because

- Before the match there is no information that “links” the skill of two players
- After the match, if the skill of the winning player goes down (e.g., due to a loss in a subsequent match) then the skill of the opponent also needs to go down (or otherwise the observed match outcome would not have been possible)

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1. Graphical Models

- Simple way to visualize the product structure of a joint probability distribution
- Useful for modelling real-life data generating processes
- Allows both to test for conditional independence and efficient marginalization (next week)

2. Bayesian Networks

- A directed acyclic graph where each edge points from a conditioning to a conditioned variable in the model
- An alternative representation (parameterization) of a joint probability (often easier to formulate for experts)
- A generative model of the data that can be easily sampled from (ancestral sampling)

3. Conditional Independence

- d-separation is a set of simple rules ("blocking") to read off conditional independence
- d-separation reduces conditional independence (exponentially hard complexity) to graph properties (polynomial complexity in sparse graphs)

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See you next week!